IGCSE 0607 Practice Questions — E4.4

Equations of linear graphs

Total Questions: 18 — **Total Marks:** 118 Marks

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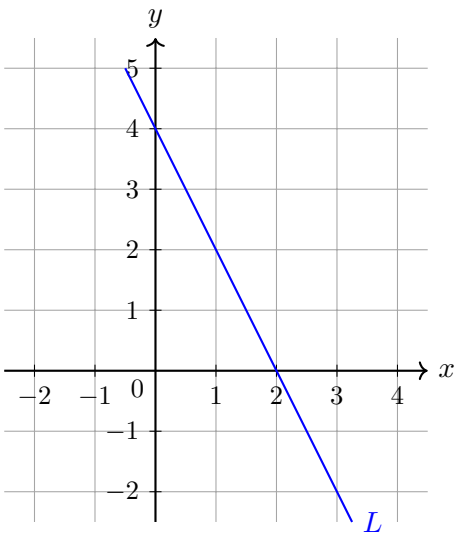
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1 Find the gradient and the y-intercept of the line with equation

$3x + 5y = 20$

Gradient = ... , y-intercept = ... **[3]**

2 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Write the equation of line L in the form $ax + by = c$, where a, b and c are integers.

..... **[1]**

3 A straight line has the equation $4x + 5y = 20$.

(a) Find the gradient and the y-intercept of this line.

Gradient = ... , y-intercept = ... **[3]**

(b) A different line is parallel to the y-axis and passes through the point $(3, -8)$.
Write down the equation of this line.

..... **[1]**

4 **(a)** Find the gradient and the y-intercept of the line $3x + 5y = 15$.

Gradient = ... , y-intercept = ... **[3]**

(b) Find the equation of the line parallel to $3x + 5y = 15$ that passes through the point $(0, -4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

5 A straight line L has the equation $3x - 4y = 12$.

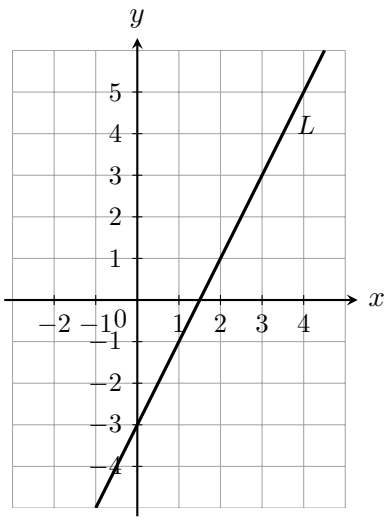
(a) Find the gradient and the y-intercept of line L.

Gradient = ... , y-intercept = ... **[3]**

(b) Another line M is parallel to L and passes through the point $(0, 5)$.
Write down the equation of line M in the form $y = mx + c$.

$y =$ ..... **[2]**

6 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the equation of line L in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) A vertical line M passes through the point $(-2, 4)$.
Write down the equation of line M.

..... **[1]**

(c) Find the coordinates of the point where line L and line M intersect.

(..... , ...) **[2]**

7 Line L_1 has equation $4x + 5y = 20$.

(a) Find the gradient and the y-intercept of line L_1.

Gradient = ... , y-intercept = ... **[3]**

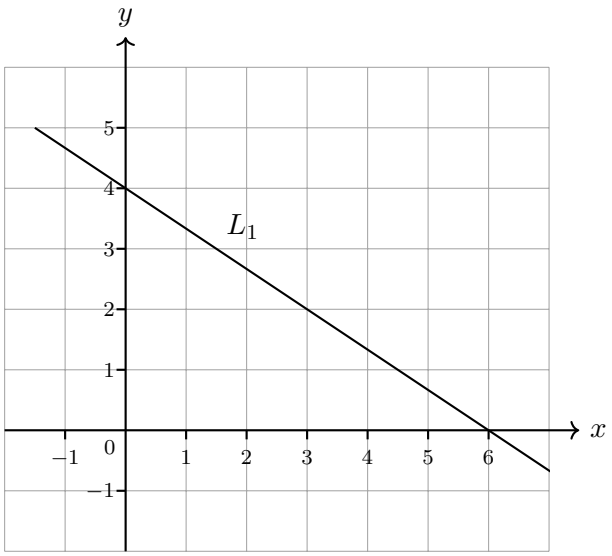
(b) Find the coordinates of the point where line L_1 crosses the x-axis.

(..... , ...) **[2]**

(c) Line L_2 passes through the points $(-3, -5)$ and $(3, 7)$.
Find the equation of line L_2 in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

8 The diagram shows a straight line L_1 drawn on a coordinate grid.



(a) Find the equation of line L_1.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

(b) The line L_2 has equation $\frac{x}{3} - \frac{y}{2} = 1$.

(i) Write the equation of L_2 in the form $y = mx + c$.

$y =$ ..... **[2]**

(ii) Find the coordinates of the point of intersection of L_1 and L_2.

(..... , ...) **[2]**

9 A is the point $(p, -1)$ and B is the point $(5, 2p - 1)$.
The gradient of the line AB is 3.

(a) Find the value of p.

$p =$ ..... **[3]**

(b) Find the equation of the line AB.
Give your answer in the form $y = mx + c$.

..... **[2]**

(c) The line AB intersects the x-axis at the point C and the y-axis at the point D.
Work out the area of triangle COD, where O is the origin.

..... **[3]**

10 A straight line L passes through the points $A(-2, 3k)$ and $B(4, k)$, where k is a non-zero constant.

(a) Find the gradient of line L in terms of k.
Give your answer in its simplest form.

..... **[2]**

(b) Find the equation of line L in the form $ax + by = c$, where a, b, and c are expressions in terms of k.

..... **[3]**

(c) Line L also passes through the point $(1, 10)$.
Find the value of k.

$k =$ ..... **[2]**

(d) Hence, write down the coordinates of the x-intercept of line L.

(..... , ...) **[2]**

11 The line L passes through the points $A(2a, a+4)$ and $B(a-1, 3a+2)$.
The gradient of line L is -4.

(a) Find the value of a.

$a =$ ..... **[3]**

(b) Find the equation of line L.
Give your answer in the form $px + qy = r$, where p, q and r are integers and $p > 0$.

..... **[3]**

(c) Line L intersects the line $y = 3x - 2$ at the point M.
Find the coordinates of M.

(..... , ...) **[3]**

12 A is the point $(-3, 1)$ and B is the point $(5, 5)$.

(a) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(b) The perpendicular bisector of AB intersects the line $x - 2y = 5$ at the point P.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Find the exact length of AP.

..... **[3]**

13 The line L passes through the points $P(-3, 10)$ and $Q(5, -2)$.

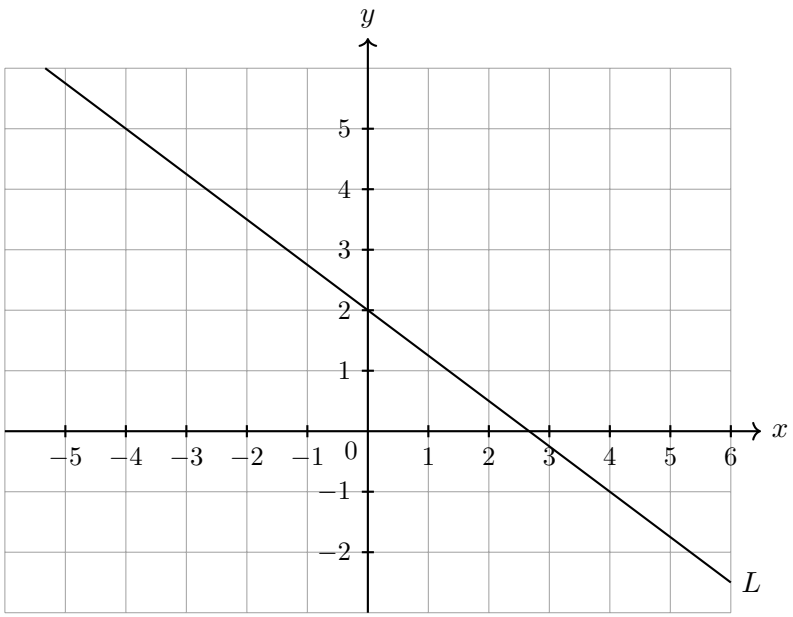
(a) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Line M has the equation $2y + 3x = 15$.
Find the coordinates of the point where line M intersects the x-axis.

(..... , ...) **[2]**

14 The diagram shows a line L drawn on a coordinate grid.



(a) Work out the gradient of line L.

..... **[2]**

(b) Find the equation of line L.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

15 Point A has coordinates $(-4, 3)$ and point B has coordinates $(6, -2)$.

(a) Find the equation of the line passing through A and B.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

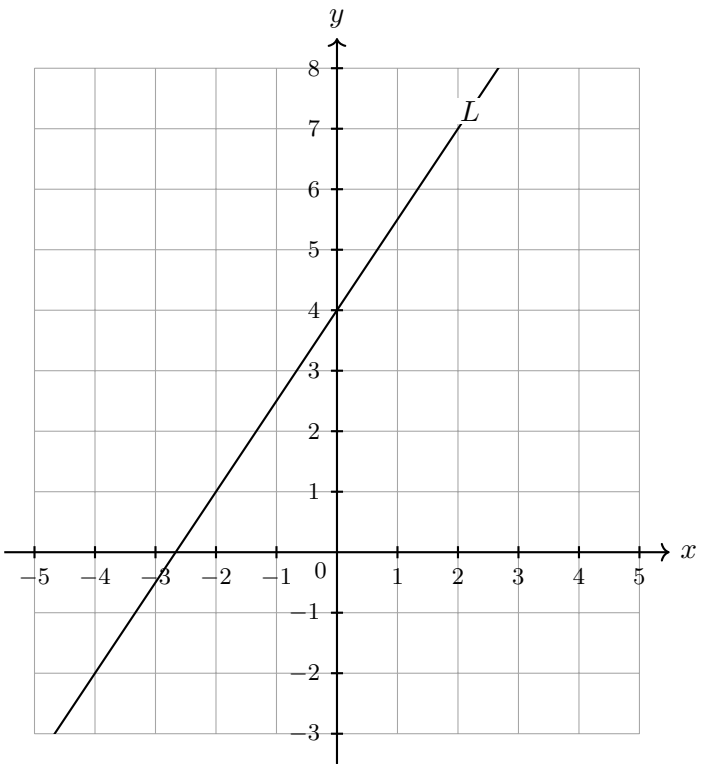
(b) The line $3x - 5y = 15$ intersects the x-axis at point C and the y-axis at point D.
Find the coordinates of C and D.

$C =$ ..... , $D =$ ..... **[2]**

(c) Write down the equation of the line that is parallel to the y-axis and passes through point A.

..... **[1]**

16 The diagram shows a straight line L.



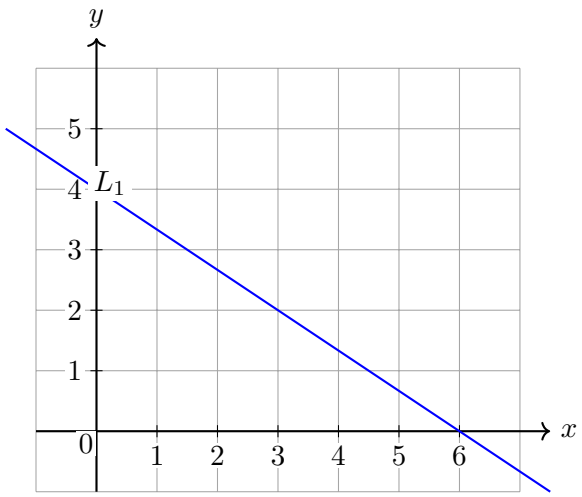
(a) Find the equation of line L.
Give your answer in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

(b) Line M is perpendicular to line L and passes through the point $(9, -1)$.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

17 The diagram shows a straight line L_1 drawn on a coordinate grid.



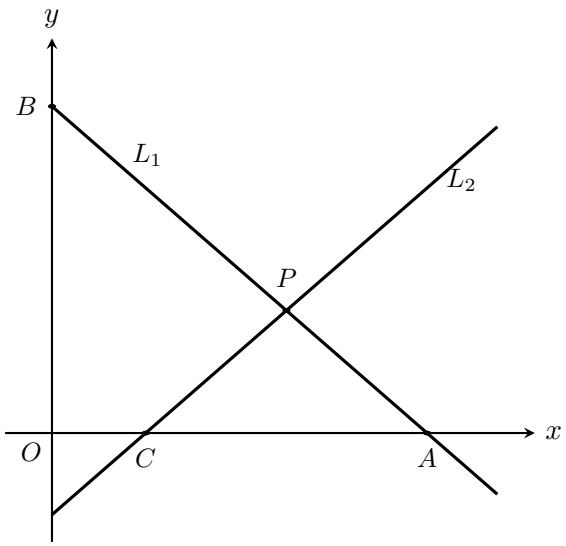
(a) Find the equation of line L_1 in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Line L_2 passes through the point $(2, 7)$ and the midpoint of the segment of L_1 that lies between the axes.
Find the equation of L_2 in the form $ax + by = c$, where a, b, and c are integers.

..... **[4]**

18 The diagram shows a sketch of two straight lines, L_1 and L_2, which intersect at the point P.
Line L_1 has equation $3x + 2y = 24$ and crosses the x-axis at A and the y-axis at B.
Line L_2 has equation $y = 1.5x - 3$ and crosses the x-axis at C.



(a) Write down the coordinates of A, B, and C.

$A =$ ..... , $B =$ ..... , $C =$ ..... **[3]**

(b) Find the coordinates of the point of intersection P.

$P =$ (..... , ...) **[3]**

(c) Calculate the area of quadrilateral $OCPB$.

..... square units **[3]**

(d) A third line L_3 has equation $kx - 4y = 14$ and also passes through point P.
Calculate the value of k.

$k =$ ..... **[2]**